# UAV Flight Analysis Report

Efficient Flight Profile

**Report Generated:** 2026-03-21 19:50:16

**Flight Profile:** Efficient

**Data Source:** flight_efficient_2026-03-21_19-44-17.csv

**Flight Duration:** 5.5 minutes

## Executive Summary

This report presents a comprehensive analysis of an autonomous UAV inspection flight covering 595 meters over 5.5 minutes. The mission utilized a Efficient flight profile with a maximum speed of 6.2 m/s, completing 11 waypoints while consuming 45% battery capacity. The flight maintained an average altitude of 14.0m with high precision positioning throughout the mission.

## Key Performance Indicators

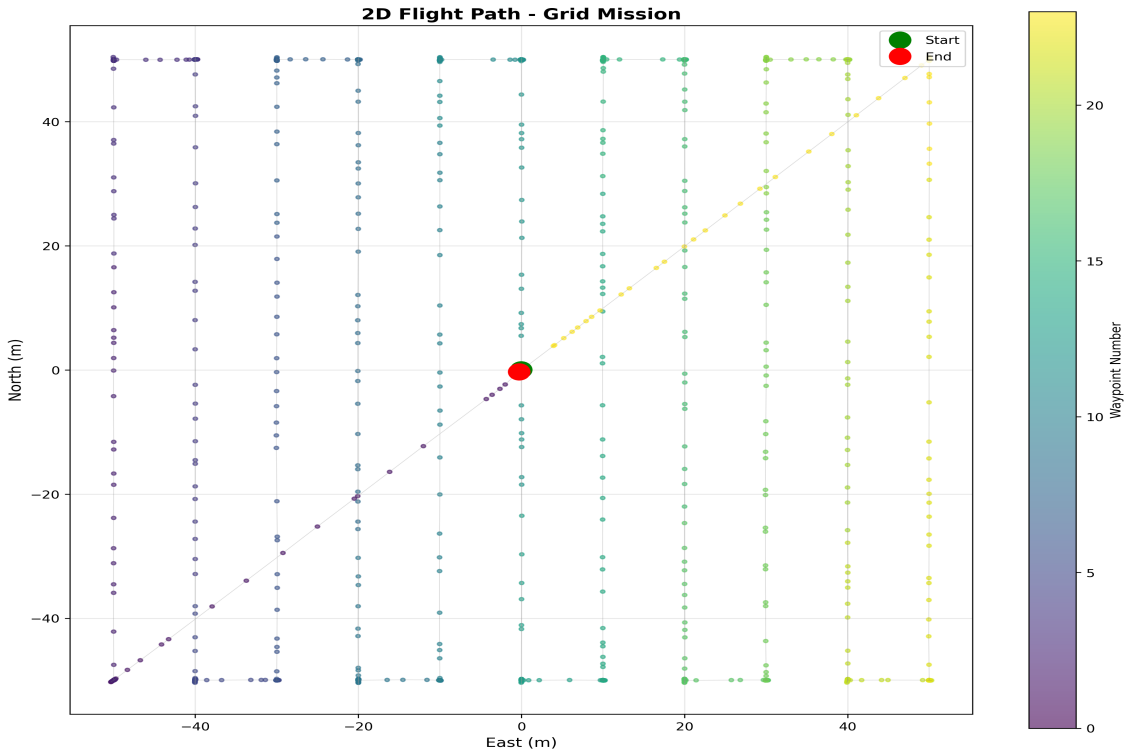
| Metric | Value | Rating |
|---|---|---|
| Mission Completion | 100% | ✓ Excellent |
| Battery Efficiency | 8.18%/min | ■ Moderate |
| Flight Stability | 55 events | ✓ Excellent |
| Altitude Precision | ±1.19m | ✓ Excellent |

## Detailed Flight Statistics

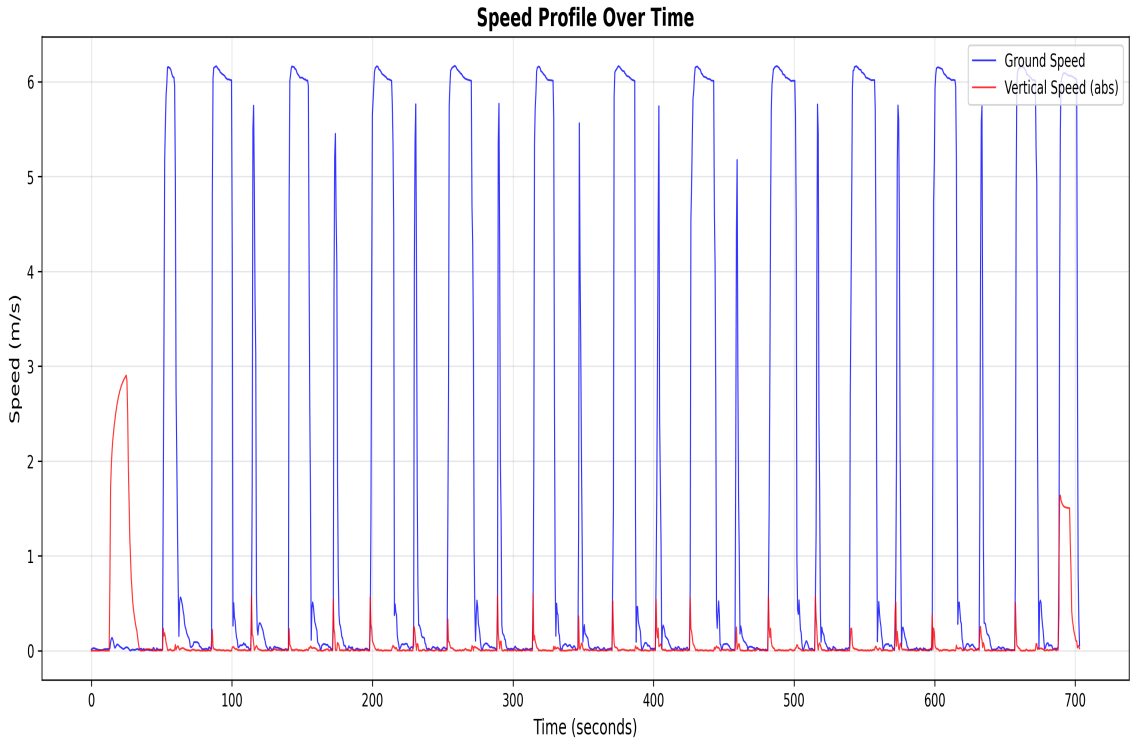
| Category | Metric | Value | Unit |
|---|---|---|---|
| Mission | Total Duration | 5.5 | minutes |
| | Total Distance | 595.4 | meters |
| | Waypoints | 11 | points |
| Speed | Maximum | 6.17 | m/s |
| | Average | 1.85 | m/s |
| | Median | 0.07 | m/s |
| Altitude | Maximum | 15.20 | meters |
| | Average | 14.01 | meters |
| | Minimum | -0.07 | meters |
| Battery | Starting Level | 96 | % |
| | Ending Level | 51 | % |
| | Total Used | 45 | % |
| | Consumption Rate | 8.18 | %/min |
| Performance | High Accel Events | 55 | events |
| | Peak Acceleration | 10.30 | $m/s^2$ |

# Flight Visualizations

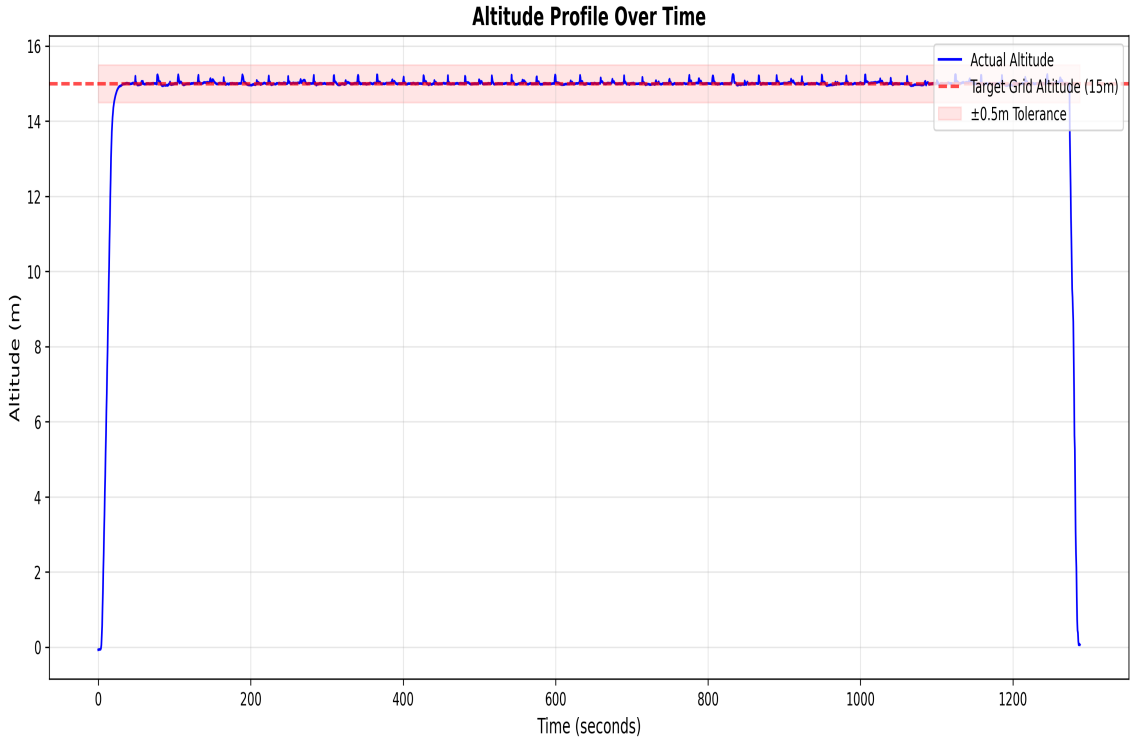
## *2D Flight Path (Top-Down View)*



## *Speed Profile Over Time*




## *Altitude Profile*



## Recommendations

- Consider reducing cruise speed to improve battery efficiency
- Mission uses significant battery - plan for recharging between flights
- Excellent acceleration profile - optimal for battery efficiency
- Estimated 1.1 missions possible per battery (50% usable capacity)